Bill of Materials (BOM) – Smart Helmet for Rider Safety (IoT PoC)

S.No	Component Name	Specification / Model	Qty	Unit Cost	Total Cost
1	Microcontroller	ESP32 Dev Board (Wi/Fi + BLE)	1	350	350
2	GSM Module	SIM800L (SMS / Call)	1	450	450
3	Alcohol Sensor	MQ3 Gas Sensor	1	150	150
4	Motion Sensor	MPU6050 (Accelerometer + Gyro)	1	120	120
5	Proximity Sensor	IR / Helmet Detection Sensor	1	60	60
6	Heart Rate Sensor	Pulse Sensor	1	250	250
7	Buzzer	Active Buzzer	1	30	30
8	Status LEDs	Red & Green LEDs	2	10	20
9	Battery	Li ion 3.7V (18650)	1	180	180
10	Charging Module	TP4056 with Protection	1	60	60
11	Boost Converter	DC DC 3.7V $\rightarrow$ 5V	1	70	70
12	Power Switch	Slide / Toggle Switch	1	20	20
13	Jumper Wires	Male–Female Set	1	50	50
14	PCB / Breadboard	Prototype Board	1	80	80
15	Enclosure / Mount	Helmet Mounting	1	50	50
				Total	1,890